# ADVANCED LINEAR ALGEBRA
# HOMEWORK 6

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**P.4.9.** If $A \in \mathbf{M}_n$ is to be represented as $A = XY^\top$ for some $X, Y \in \mathbf{M}_{n\times r}$, explain why $r$ cannot be smaller than $\operatorname{rank}(A)$.

*Solution.* We claim that the columns of $A$ must lie in the column space of $X$. To see this we write

$$Y^\top = \begin{bmatrix} y_1^\top \\ y_2^\top \\ \vdots \\ y_r^\top \end{bmatrix},$$

where $y_i \in \mathbb{F}^n$ are columns of $Y$. Then we have that

$$XY^\top = \sum_{k=1}^{r} x_k y_k^\top.$$

Now choose an arbitrary vector $v \in \mathbb{F}^n$,

$$Av = XY^\top v = X(Y^\top v)$$

with $Y^\top v \in \mathbb{F}^r$. So $Av$ is a linear combination of the $r$ columns of $X$, which gives us $\operatorname{col}(A) \subseteq \operatorname{col}(X)$. So

$$\operatorname{rank}(A) = \dim(\operatorname{col}(A)) \leq \dim(\operatorname{col}(X)) = \operatorname{rank}(X) \leq r$$

So $\operatorname{rank}(A) \leq r$, and we are done. $\square$

**P.4.10.** For $X \in \mathbf{M}_{m\times n}$, $\nu(X)$ denote the nullity of $X$.

(a) Show that $\nu(X^\top) = \nu(X) + m - n$.
(b) Let $A \in \mathbf{M}_{m\times k}$ and $B \in \mathbf{M}_{k\times n}$. Show that the rank inequality given by

$$\operatorname{rank}(A) + \operatorname{rank}(B) - k \leq \operatorname{rank}(AB)$$

is equivalent to

$$\nu(AB) \leq \nu(A) + \nu(B). \tag{1}$$

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(c) If $A, B \in \mathbf{M}_n$, show that the rank inequality given by

$$\operatorname{rank}(AB) \leq \min\{\operatorname{rank}(A), \operatorname{rank}(B)\}.$$

is equivalent to

$$\max\{\nu(A), \nu(B)\} \leq \nu(AB) \tag{2}$$

The inequalities (**??**) and (**??**) are known as *Sylvester's law of nullity.*

(d) Consider $A = [0 \quad 1]$ and $B = [1 \quad 0]^\top$. Show that the inequality (**??**) need not be valid if the matrices are not square.

*Proof.* □

**P.4.13.** Let $\mathcal{V}$ be a nonzero $n$-dimensional vector space over the field $\mathbb{F}$, and let $T \in \mathfrak{L}(\mathcal{V}, \mathbb{F})$ be a nonzero linear transformation. Why is $\dim(\ker(T)) = n - 1$

*Solution.* □

**P.4.35.** For each of the following matrices, compute its reduced row echelon form, its pivot column decomposition, and bases for its column space and null space.

$$\text{(a)} \quad \begin{bmatrix} -1 & -2 & 2 & 3 & 3 \\ 3 & 6 & 1 & 5 & -1 \\ 2 & 4 & -1 & 0 & 2 \end{bmatrix} \qquad \text{(b)} \quad \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 1 & 0 & 1 & 1 & 1 \\ 3 & 2 & 2 & 3 & 2 \end{bmatrix}.$$

*Solution.* □

**P.4.38.** Suppose that $A \in \mathbf{M}_{6\times 4}$ is row equivalent to

$$B = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 3 & 2 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(a) Find a basis for $\operatorname{null}(A)$.
(b) What is $\dim(\operatorname{null}(A^\top))$

*Solution.* □